a) **Subunit recombinant vaccines**
b) attenuated recombinant vaccines
c) DNA vaccines
d) conventional vaccines

10. Mention the number of primers required in each cycle of PCR. Write the role of primers and DNA polymerase in PCR. Name the source organism of the DNA polymerase used in PCR.
11. How is the amplification of a gene sample of interest carried out using PCR?
12. What is genetically engineered Insulin?
13. Explain how "Rosie" is different from a normal cow.
14. How was Insulin obtained before the advent of rDNA technology? What were the problems encountered?
15. ELISA is a technique based on the principles of antigen-antibody reactions. Can this technique be used in the molecular diagnosis of a genetic disorder such as Phenylketonuria?
16. Gene therapy is an attempt to correct a Genetic defect by providing a normal gene into the individual. By this the function can be restored. An alternate method would be to provide gene product known as enzyme replacement therapy, which would also restore the function. Which in your opinion is a better option? Give reasons for your answer.
17. What are transgenic animals? Give examples.
18. If a person thinks he is infected with HIV, due to unprotected sex, and goes for a blood test. Do you think a test such as ELISA will help? If so why? If not, why?
19. Explain how ADA deficiency can be corrected?
20. What are DNA vaccines?
21. Differentiate between Somatic cell gene therapy and germline gene therapy.
22. What are stem cells? Explain its role in the field of medicine.
23. One of the applications of biotechnology is 'gene therapy" to treat a person born with a hereditary disease.
    i) What does "gene therapy" mean?
    ii) Name the hereditary disease for which the first clinical gene therapy was used.
    iii) Mention the steps involved in gene therapy to treat this disease.
24. PCR is a useful tool for early diagnosis of an Infectious disease. Elaborate.
25. What are recombinant vaccines?. Explain the types.
26. Explain why cloning of Dolly, the sheep was such a major scientific breakthrough?
27. Mention the advantages and disadvantages of cloning.
28. Explain how recombinant Insulin can be produced.

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